

Enhanced Model Predictive Control of Voltage Profiles in MV Networks with Distributed Generation

Gain Scheduling, Rate Constraints, and Node-Wise Soft Constraints

Rayana Kristina Schneider Barcelos - Federal Institute of Espirito Santo (IFES), Sao Mateus, ES, Brazil

ABSTRACT

This paper extends the impulse-response model predictive control framework proposed by Farina et al. for voltage regulation in medium-voltage distribution networks with distributed generation. The baseline controller is augmented with a bank of FIR models selected by gain scheduling, penalties and hard bounds on input-rate variations, node-wise soft voltage constraints, and predictive OLTC supervision. Because the original publication does not provide the numerical impulse-response matrices used in its PowerFactory case study, the comparison uses a reproducible surrogate FIR benchmark preserving the original dimensions and principal tuning parameters. The enhanced controller reduces maximum voltage-limit violation by 21.6%, the integral of violations by 40.2%, node-seconds outside the band by 32.8%, total power-factor variation by 40.9%, and the largest power-factor step by 79.1%, without increasing OLTC operations.

I. INTRODUCTION

High penetration of distributed generation can produce local voltage rise, reverse power flow, and demanding voltage-control problems in radial medium-voltage networks. Farina et al. proposed a hierarchical architecture in which a static optimal power-flow layer generates voltage and reactive-power references, a centralized MPC layer computes distributed-generator power-factor references, and local automatic voltage regulators track reactive-power commands. Their approach uses an input-output impulse-response model, avoids explicit state estimation, and handles voltage and control constraints directly in a quadratic program. The original study reports oscillatory behavior under operating-point mismatch and suggests gain scheduling, multiple linearized models, and more slack variables as future improvements. This work implements these directions and adds explicit rate penalties and rate constraints.

II. BASELINE MPC FORMULATION

The baseline model is $y(k) = \sum_i [g_i u(k-i) + \gamma_i d(k-i)] + \delta(k)$, where y contains voltage deviations, u contains distributed-generator power-factor references, d contains measurable disturbances, and δ summarizes unknown exogenous contributions. The original benchmark uses $T=2$ s, $M=90$, eleven controlled voltages, eight manipulated power factors, $N=10$, $N_u=2$, $Q=10$ I, $R=0.1$ I, voltage limits 0.9 to 1.1 pu, and power-factor limits 0.6 to 1.0. At each sample the quadratic program minimizes tracking error, control effort, and soft-constraint violations. Only the first optimal move is applied.

III. ENHANCED CONTROL STRATEGY

A. Scheduled FIR model bank. Multiple local FIR models are stored and an operating-region indicator selects the model

that best matches loading and generation. B. Input-rate regularization. The cost adds $\Delta u' R \Delta u$ and hard bounds $|\Delta u| \leq \Delta u_{\max}$, suppressing rapid power-factor changes. C. Node-wise soft constraints. Separate lower and upper slacks are assigned to each controlled voltage and prediction step: $V_{\min} - \epsilon_{\text{low}} \leq V_{\text{pred}} \leq V_{\max} + \epsilon_{\text{high}}$. D. Predictive OLTC supervisor. When persistent slack activation indicates insufficient reactive-power authority, candidate tap positions are evaluated and the action that most reduces predicted aggregate violation is selected, subject to a dwell-time condition.

IV. REPRODUCIBLE COMPARATIVE STUDY

Exact numerical reproduction of the original PowerFactory

impulse-response matrices g_i and γ_i are not published. The baseline and enhanced algorithms are therefore compared on a reproducible surrogate FIR plant with the same primary dimensions and tuning values as the original controller. The surrogate includes operating-point changes and large disturbances to stress model mismatch and constraint handling. Both controllers use identical plant trajectories, voltage limits, power-factor limits, prediction horizon, and control horizon.

V. RESULTS

Maximum voltage violation falls from 0.03686 pu to 0.02891 pu (21.6%). The integral of voltage violations falls from 4.225 to 2.527 pu*s (40.2%), while node-seconds outside the band fall from 458 s to 308 s (32.8%). Mean absolute voltage deviation from 1 pu falls by 8.5%. Total power-factor variation falls from 22.38 to 13.23 (40.9%), and the maximum individual power-factor step falls from 0.382 to 0.080 (79.1%). Both controllers perform five OLTC operations. Minimum voltage changes from 0.9311 to 0.9276 pu and remains above 0.9 pu, while maximum voltage falls from 1.1369 to 1.1289 pu.

VI. DISCUSSION

The results support using scheduled local models when the operating regime changes. Explicit rate shaping is preferable to post-filtering because it is incorporated into the optimization and respects constraints by construction. Node-wise slacks provide richer diagnostic information than two global slack variables and make OLTC coordination more transparent. The principal limitation is the surrogate plant; the reported percentages are benchmark results rather than a reproduction of the original Italian MV network.

VII. CONCLUSIONS

The enhanced impulse-response MPC preserves the computationally attractive quadratic-program structure while improving robustness to operating-point changes, voltage-constraint management, and control smoothness.

Future work should validate the approach on a full network model, compare fixed, gain-scheduled, and adaptive FIR MPC under identical conditions, and investigate distributed implementations.

REFERENCES

[1] M. Farina et al., "Model predictive control of voltage profiles in MV networks with distributed generation," 2013. [2] E. F. Camacho and C. Bordons, Model Predictive Control, Springer, 2007. [3] R. Scattolini, "Architectures for distributed and hierarchical model predictive control - a review," Journal of Process Control, vol. 19, pp. 723-731, 2009.